Top Picks Recommendation Score

HealthyFoodsNYC (team-3-fall26) · branch `feature/recommend-score` · Jonghyun Jeong · 2026-11-07 · final version v3

1. Summary

Final score (0–100) = 100 × (health × 50% + inspection grade × 25% + rating × 25%) + profile adjustments
With **BMI 25+** (or a weight-loss goal) the weights become **health 60% / grade 25% / rating 15%**. BMI is never shown on the page.
Profile adjustments: allergy-risky cuisines removed, favorite food +10, disliked food −15, near home +5
The three highest scores are shown as Top picks, each card with up to three reason tags. Price is not part of the score; it only appears as the Budget-friendly tag.

The score is a rule-based weighted sum, so every result can be explained part by part. It is computed by `score_restaurant()` in `search/recommend.py`, which returns `(score, reason tags)`.

2. Problems and Fixes

	Problem	Fix
1	Using Google ratings and reviews as-is surfaces **famous, tasty places**, such as a $1 pizza shop, not healthy ones.	The **health score** (cuisine + review content) gets the largest weight, **50%**; rating gets 25%. Reviews are read for health-related content, not for their stars.
2	With price in the score, **fast food ranked first** because most chains are `$` (v2: McDonald's 75, salad chains 66).	**Price was removed from the score** and is shown only as the Budget-friendly tag.
3	Cuisine alone does not catch all fast food. Taco Bell is listed as Tex-Mex, so it was not penalized.	A **list of fast-food chains** (23 names) lowers the health score by 0.2.
4	Health matters more for some users than others.	With **BMI 25+** or a **weight-loss goal**, the health weight rises to 60%. BMI is never shown on the page.

3. Score Weights

Part	Default	Health focus BMI 25+ or weight-loss	What it measures
Health	50%	60%	Cuisine, share of healthy reviews, fast-food chain
Inspection grade	25%	25%	Latest NYC health inspection grade
Rating	25%	15%	Google rating blended with in-app ratings
Price	0%	0%	Not scored; shown as a tag only

4. How Each Part Is Calculated

4-1. Health score (0–1)

```
health = 0.5
       + 0.2  healthy cuisine       / − 0.2  unhealthy cuisine
       + 0.3 × (share of healthy reviews − share of unhealthy reviews)
       − 0.2  fast-food chain
       (clipped to 0–1)
```

Item	Rule
Healthy cuisines	Salads, Vegetarian, Vegan, Juice/Smoothies, Fruits/Vegetables, Soups, Mediterranean, Greek, Middle Eastern, Japanese, Korean, Seafood, Hawaiian
Unhealthy cuisines	Hamburgers, Hotdogs, Chicken, Pizza, Donuts, Frozen Desserts, Bakery Products/Desserts, Nuts/Confectionary
Healthy review	A review that uses more healthy words (healthy, fresh, salad, grilled, steamed, organic, light, vegan, ...) than unhealthy ones. Each review counts once, however many words it uses.
Unhealthy review	A review with more unhealthy words (greasy, oily, fried, heavy, salty, sugary, processed).
Review source	Up to 5 reviews that Google picks by relevance; saved reviews in our DB when Google has none.
Fast food	Names starting with McDonald's, Burger King, Wendy's, Taco Bell, Popeyes, KFC, Domino's, Papa John's, Pizza Hut, Wingstop, Five Guys, Chick-fil-A, Dunkin', etc. (23 chains). Subway, Chipotle and Sweetgreen are not on the list.

4-2. Inspection grade (0–1)

A = 1.0 · B = 0.5 · C = 0 · pending (P, Z, N) or no grade = 0.3

4-3. Rating (0–1)

```
blended = (google_rating×google_count + app_avg×app_count + 4.0×20) ÷ (google_count + app_count
+ 20)
rating  = (blended − 3.0) ÷ 2.0   (0 at 3.0 or below, 1 at 5.0)
```

Places with few votes are pulled toward an average of 4.0, so two 5-star votes cannot beat 4.7 stars from 500 reviews.

4-4. Profile adjustments (logged-in users with a profile)

Profile field	Effect
Allergies	Cuisines that often use nuts or shellfish are **removed** from the candidates (existing diet_filters).
Favorite foods	Whole-word match on cuisine or name: **+10**, Your favorite tag
Disliked foods	Match: **−15**

Home address	Same ZIP or borough: **+5**, Nearby tag
Health goals	Add goal words to the healthy-review check (e.g. muscle-gain adds protein, chicken).
BMI (weight / height)	If `703 × lbs ÷ inches²` is **25+** (or there is a weight-loss goal), weights switch to **health 60 / grade 25 / rating 15**. Never shown on the page (section 5).

5. How BMI Is Used

- The profile stores **pounds and inches**, not kg and cm: `BMI = 703 × lbs ÷ inches²`
- With **BMI 25+** or a **weight-loss** goal, the health-focus weights apply (health 60, grade 25, rating 15): 10 points move from rating to health.
- If weight or height is missing, the default weights apply.
- **BMI is never shown on the page or in a tag.** It only changes the weights.
- BMI cannot tell muscle from fat, so it is combined with the user's own weight-loss goal.

Example (v3 code, 5 made-up restaurants, all grade A, height 68 in)

Restaurant	Cuisine	Google rating (count)	Health	Rating part	150 lb BMI 22.8	200 lb BMI 30.4
Poke Place	Hawaiian	4.0 (200)	0.90	0.50	82	86
Ramen Bar	Japanese	4.7 (500)	0.70	0.84	81	80
Taqueria	Mexican	4.6 (400)	0.60	0.79	75	73
Green Bowl	Salads	3.2 (100)	0.90	0.17	74	82
Slice House	Pizza	4.8 (900)	0.30	0.89	62	56

At BMI 22.8 the top 3 are Poke, Ramen and Taqueria. At BMI 30.4 Taqueria (**75→73**) drops out and Green Bowl, healthy but lower rated, moves to 2nd (**74→82**).
Green Bowl: default $0.9 \times 50 + 25 + 0.17 \times 25 = 74.2$ → health focus $0.9 \times 60 + 25 + 0.17 \times 15 = 81.6$

6. Reason Tags

Tag	Rule
Your favorite	Matches a favorite food
Nearby	Same ZIP or borough as the home address
Healthy reviews	At least 2 healthy reviews, and at least 40% of all reviews
Budget-friendly	Google price is `$` (no effect on the score)
Top rated	50+ votes and a blended rating of 4.5+
Grade A	Latest inspection grade is A

Tags are shown in this order, at most three per card. Student pick was removed in v3 because we do not check whether a user is a student.

7. Candidate Flow

1. Take the 9 most recent grade-A restaurants. Cuisines risky for the user's allergies are excluded up front.
2. For logged-in users, add up to 3 matching a favorite food and up to 3 in the home borough (15 at most).
3. Attach Google Places data (price, rating, 5 reviews), the NYC inspection record, and in-app ratings.
4. Score them and show the top 3. Top picks appear only on the first page with no borough selected.

8. Versions and Real Results

Version	Weights (price/health/grade/rating)	Why it changed
v0 (before)	–	Showed the 3 most recent grade-A restaurants by date.
v1	40 / 25 / 20 / 15	First design, built for students, so price had the largest weight.
v2	40/25/20/15, health focus 30/35/20/15	Added BMI and the weight-loss goal.
v3 (final)	0 / 50 / 25 / 25, health focus 0/60/25/15	Fast food ranked first, so price was removed and review shares plus a fast-food penalty were added.

v1: from date order to score order (real local data)

Top 3: NAM PO BK 83 · CAFE AROMA Y SABOR 82 · AGAVE & MAIZ 79 (Student pick Healthy reviews). TACO BELL, which was in the top 3 by date, dropped out. This run had no logged-in profile.

v2: BMI (real local data, after re-importing cuisine data)

Rank	BMI 22.8 (150 lb)	BMI 30.4 (200 lb)
1	CAFE AROMA Y SABOR (Coffee/Tea, $) 85	AGAVE & MAIZ (Mexican, $$) **82**
2	NAM PO BK (Asian, $) 83	CAFE AROMA Y SABOR **81**
3	AGAVE & MAIZ 79	NAM PO BK **79**

The change is roughly 10 × (health − price part). Agave & Maiz has many healthy reviews (health ≈ 0.9), so it gained 3 points while the other two lost 4, moving it from 3rd to 1st. No salad places were among the candidates, so only the order changed.

v2 check: fast food on top (real Google data)

Restaurant	Cuisine	Price	Score
McDonald's	Hamburgers	$	75
Popeyes	Chicken	$	74

Wingstop	Chicken	$	70
Burger King / Wendy's / KFC / Domino's	Hamburgers · Chicken · Pizza	$	69
Sweetgreen / Just Salad	Salads	–	66
Chopt / Dig Inn	Other · American	–	60
Taco Bell	Tex-Mex	–	58

The cause was price, not inspection grade. McDonald's 75 = price 1.0×40 + health 0.35×25 + grade 1.0×20 + rating 0.4×15 = 40 + 8.75 + 20 + 6. The unhealthy-cuisine penalty is worth about −6 points in the final score, but the gap between `$` and `$$` was 16 points. Taco Bell, listed as Tex-Mex, had no penalty at all.

v3: no price, review shares, fast-food penalty (real Google data)

Restaurant	Cuisine	Health	Score
Sweetgreen / Just Salad	Salads	0.70	72
Chopt / Dig Inn	Other · American	0.50	62
Taco Bell	Tex-Mex	0.30	50
McDonald's	Hamburgers	0.16	43
Popeyes	Chicken	0.10	42
Wingstop	Chicken	0.10	36
Burger King / Domino's	Hamburgers · Pizza	0.10	35
Wendy's / KFC	Hamburgers · Chicken	0.10	34

- McDonald's health 0.16 = 0.5 − 0.2 (Hamburgers) − 0.2 (fast food) + 0.3 × 1/5 (1 of 5 reviews healthy); score 75 → **43**
- Taco Bell health 0.5 → 0.3 (now detected as fast food); score 58 → **50**
- Salad chains 66 → **72**, computed without reviews because the Google match failed; a match would raise them further.

9. Why Calorie Data Is Not in the Score

Our DB has DOHMH MenuStat calorie data (`MenuItem`, about 65k menu items, 85% with calories). We considered using it to recommend restaurants but left it out of the score for these reasons.

Problem	Details
Large chains only	Menus from 96 national chains (McDonald's, Starbucks, Subway, Chipotle, ...). No data for local restaurants.
Only about 4% of restaurants link	There is no shared ID, so names must match. Of about 31.5k inspected restaurants, about 1,300 (46 chains) match by name. The other 96% cannot be judged on calories, which makes the comparison unfair.

Mostly drinks and toppings	About 23k items are beverages and about 10k are toppings or sauces. Sorting by calories puts Vinegar and Diet Coke (0 kcal) on top.
Old data	Menus are from 2017–2018 and may differ from today's.
Little effect on Top picks	Chains rarely appear among the candidates (9–15 recent grade-A places), and those that do are mostly fast food, already penalized.

Example: Subway menu

Sorted by calories only	kcal	Meals only + protein 15 g+	kcal / protein
Vinegar (topping)	0	Roast Beef Salad	140 / 19 g
Diet Coke (drink)	0	Subway Club Salad	140 / 18 g
Fuze Unsweetened Tea (drink)	0	Oven Roasted Chicken Salad	140 / 19 g
Sauerkraut (topping)	10	Carved Turkey Salad	160 / 20 g

Decision: calories stay out of the score. A separate PR will show a **top-5 low-calorie menu** on chain restaurants' detail pages, using meal categories only (Entrees, Sandwiches, Salads, Burgers, Pizza, Soup), protein 15 g+, no kids' meals, latest year, one row per item name. Chain names match reliably, so the data is accurate for this use.

10. Verification

- All 191 tests pass; black 26.10.1 and flake8 7.4.1 pass (same as CI).
- New `search/test_recommend.py`: weights sum to 1.0 with no price; each review counts once; fast-food detection (Taco Bell, KFC/TACO BELL, CHICK-FIL-A yes, THE TACO SHOP no); under equal conditions Sweetgreen beats McDonald's by 20+ points; BMI 25+ flips the order; allergy exclusion; in-app ratings count; at most 3 tags.
- Google match rate: 7 of 9 Top picks candidates (78%).

11. Limits and Next Steps

Limit	Next step
Google gives at most 5 reviews per place, chosen by relevance.	Include our community comments in the review shares.
Health is keyword-based.	Discuss with the team replacing only the health-keyword part with an LLM.
Renamed or food-hall locations (Dig Inn, Chopt) fail the Google match.	Track the matching rules as a separate issue.
Calorie data (MenuStat) covers 96 chains, about 4% of restaurants.	Keep it out of the score; build a separate PR for a top-5 low-calorie menu on the detail page (meals only, protein 15 g+).
About 12% of restaurants have no cuisine.	Their health score uses reviews and the fast-food check only.

Each candidate needs a Google call: 9–15 calls on first load.	Lower POOL_SIZE if cost becomes a problem.